A FILTERED COLIMIT OF SMOOTH ALGEBRAS WITH NOETHERIAN FIBRES IS A REGULAR RING MAP

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness of colimits and of smooth algebras. Algebra, Lemma \ref{algebra-lemma-colimit-flat} and Algebra, Lemma \ref{algebra-lemma-smooth-syntomic}, cited at proof lines 10466-10467. Give flatness of the colimit over the base.
- Base change of smooth algebras is smooth. Algebra, Lemma \ref{algebra-lemma-base-change-smooth}, cited at proof lines 10474-10476. Identifies the base changed fibre as a colimit of smooth algebras over the extended field.
- Local rings of smooth algebras over a field are regular, and regularity is preserved by filtered colimits. Algebra, Lemma \ref{algebra-lemma-characterize-smooth-over-field} and Algebra, Lemma \ref{algebra-lemma-colimit-regular}, cited at proof lines 10477-10482. Prove regularity of the local rings of the base changed fibre.

Source reference. Original lemma 07EP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9454–9454. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05UT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7527–7527. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00TA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1068–1068. Navigation pointer to a statement the

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Source reference. Original lemma 00T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1062–1062. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00TT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1087–1087. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07DX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9427–9427. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let $(A_i, \varphi_{ii'})$ be a directed system of smooth R-algebras. Set $\Lambda = \operatorname{colim} A_i$. If the fibre rings $\Lambda \otimes_R \kappa(\mathfrak{p})$ are Noetherian for all $\mathfrak{p} \subset R$, then $R \to \Lambda$ is regular.*

Proof. Note that Λ is flat over R by Algebra, Lemmas 05UT and 00TA. Let $\kappa(\mathfrak{p}) \subset k$ be a finite purely inseparable extension. Note that

$$\Lambda \otimes_R \kappa(\mathfrak{p}) \otimes_{\kappa(\mathfrak{p})} k = \Lambda \otimes_R k = \operatorname{colim} A_i \otimes_R k$$

is a colimit of smooth k-algebras, see Algebra, Lemma 00T4. Since each local ring of a smooth k-algebra is regular by Algebra, Lemma 00TT we conclude that all local rings of $\Lambda \otimes_R k$ are regular by Algebra, Lemma 07DX. This proves the lemma. □